EPS 704 Week 5: Exploratory Analysis and Statistics

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https://github.com/jxd1586/JennaD

```
library(readr); library(dplyr); library(tidyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
enroll <- read_csv("enrollment.csv")
```

```
## Rows: 800 Columns: 25

## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr   (8): student_id, program_code, program, degree_level, enrolled_status,...
## dbl  (16): cohort_year, graduated, age, international, gpa, credits, gre_q, ...
## date  (1): enrolled_on
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
enroll <- enroll |>
  mutate(gpa = na_if(gpa, 999))
```

```
library(readr)
enroll <- read_csv("enrollment.csv")
```

```
## Rows: 800 Columns: 25
## -- Column specification --------------------------------------------------------
## Delimiter: ","
```

```
## chr   (8): student_id, program_code, program, degree_level, enrolled_status,...
## dbl  (16): cohort_year, graduated, age, international, gpa, credits, gre_q, ...
## date  (1): enrolled_on
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
cor.test(enroll$gpa, enroll$credits)
```

```
##
##  Pearson's product-moment correlation
##
## data:  enroll$gpa and enroll$credits
## t = -1.0077, df = 798, p-value = 0.3139
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.10470442  0.03374857
## sample estimates:
##         cor
## -0.03564899
```

```r
## The result shows a weak negative relationship with gpa and credits and the p value is not statistica
t.test(gpa ~ graduated, data = enroll)
```

```
##
##  Welch Two Sample t-test
##
## data:  gpa by graduated
## t = 0.63487, df = 549.98, p-value = 0.5258
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
##  -18.31833  35.81435
## sample estimates:
## mean in group 0 mean in group 1
##        43.44577        34.69776
```

```r
## The t-test p value is not statistically significant.
summary(aov(gpa ~ program, data = enroll))
```

```
##              Df   Sum Sq Mean Sq F value Pr(>F)
## program       5    45682    9136   0.254  0.938
## Residuals   794 28579874   35995
```

```r
## The F value and the p value show there is no statistical significance between the relationship of gp
chisq.test(table(enroll$degree_level, enroll$graduated))
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  table(enroll$degree_level, enroll$graduated)
## X-squared = 0.029667, df = 1, p-value = 0.8632
```

```r
## The p value again shows no relationship between degree level and graduated.
summary(lm(gpa ~ credits, data = enroll))
```

```
##
## Call:
## lm(formula = gpa ~ credits, data = enroll)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -49.17 -43.05 -36.33 -31.12 969.06
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   65.536     25.569   2.563   0.0106 *
## credits       -2.373      2.355  -1.008   0.3139
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 189.3 on 798 degrees of freedom
## Multiple R-squared:  0.001271,   Adjusted R-squared:  1.931e-05
## F-statistic: 1.015 on 1 and 798 DF,  p-value: 0.3139
```

```r
## The p value shows no statistical significance between credits and gpa.
pp <- read_csv("pre_post_scores.csv")
```

```
## Rows: 120 Columns: 3
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (1): student_id
## dbl (2): pre, post
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
t.test(pp$post, pp$pre, paired = TRUE)
```

```
##
##  Paired t-test
##
## data:  pp$post and pp$pre
## t = 7.3211, df = 119, p-value = 3.169e-11
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
##  3.052494 5.315839
## sample estimates:
## mean difference
##        4.184167
```

```r
## The p value is extremely small and shows a significant statistical difference between the pre and po
```